Junaid Ali Bhatti

Senior AI Engineer | Full-Stack Developer

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Summary

Senior AI engineer with 4 years of experience building production LLM agent systems and the platforms that run them — LangGraph and LangChain workflows, MCP integrations, RAG over vector databases, model-agnostic LLM integration, multi-tenant FastAPI backends, React/Next.js front ends, deployed on AWS. MS thesis contributes a novel GAN architecture that outperforms 14 published baselines, so I build models as well as integrate them.

Work Experience

Senior AI Engineer – Techesthete, Lahore *Oct 2026 – Present*

- Build and deploy **AI agents** and agentic workflows with **LangChain** and **LangGraph**, integrating **OpenAI**, **Claude** and **Gemini** models.
- Develop LLM-powered applications and **RAG** workflows over vector databases, from ingestion to grounded responses.
- Build **MCP** integrations that give LLM agents controlled access to tools and data.
- Develop **Python** and **Node.js** backend services and **React.js** AI interfaces, and deploy them to production.

Software Engineer – Smart A and I Solutions PVT Ltd, Lahore *Sep 2025 – Sep 2026*

- Led development of two AI products from inception — planning, work breakdown, code review, Git workflow, and the hardest technical components personally.
- Architected multi-tenant AI SaaS end-to-end on **FastAPI** (async SQLAlchemy, PostgreSQL) and **Next.js**.
- Built agentic LLM systems with **LangGraph**, **Claude** and **OpenAI** — lead qualification, document intelligence, citation-grounded RAG.
- Owned deployment on **AWS**: EC2, RDS, S3, IAM roles and policies, Secrets Manager, and CI/CD pipelines.

Software Developer – Vital Green PVT Ltd, Lahore *Jan 2023 – Aug 2025*

- Founding developer on **CropSight**, one of three engineers who took it from prototype to a deployed agri-industry platform.
- Moved into the technical lead role as the team grew to nine, directing architecture and onboarding new developers.
- Built the web application and owned deployment, server administration and task automation on Ubuntu.

Full Stack Developer (Contract) – Logic Finder, Reston, Virginia, US *Mar 2025 – Jun 2025*

- Built **Network Fort**, a real-time cybersecurity dashboard, using **Next.js** and Server-Sent Events.
- Designed secure **REST APIs** with **Elasticsearch** integration; worked directly with the US client team.

Lab Engineer – University of Management and Technology (UMT), Lahore *Jun 2024 – Jul 2025*

- Conducted labs for programming and modern tech trends within the School of Systems and Technology.

Research Associate – COMSATS University Islamabad, Lahore Campus *Feb 2024 – Jun 2025*

- Conducted lab sessions for C++ Programming, Operating Systems, and AI, and guided student projects and research.

Selected Projects

PropAI — WhatsApp-First AI Platform for Real Estate Agents *Live*

FastAPI, PostgreSQL + pgvector, Redis/ARQ, Claude API, OpenAI embeddings, Next.js, Stripe, AWS

- **Lead developer** — multi-tenant SaaS where agents run their entire lead, buyer and listing pipeline through WhatsApp.
- Claude-driven lead qualification and daily agent briefings; semantic buyer-to-listing matching via pgvector.
- Tenant-scoped RBAC, 2FA, Playwright listing ingestion, and Stripe usage metering.

ELENA — Private Document Intelligence & AI Chat Platform *Live*

Python 3.12, FastAPI, async SQLAlchemy, PostgreSQL, Next.js 16, React 19, TypeScript, Docker

- **Lead developer** — RAG platform with citation grounding; answers resolve to source chunks, with web-search fallback on empty retrieval.
- Model-agnostic LLM service layer built for pluggable providers and future self-hosted models.
- Persistent cross-session memory system and in-chat document mentions.

Marginalia — Chat-With-Your-PDF Study Platform *Live* | *GitHub*
Next.js 14, TypeScript, PostgreSQL + pgvector, Prisma, Inngest, Vercel AI SDK, OpenAI/Gemini

- RAG app where users upload a PDF and get page-cited answers (each citation opens the PDF at that page), plus generated summaries, flashcards and quizzes.
- Durable **Inngest** pipeline: parse pages, chunk with page numbers, embed into pgvector, then generate study material, with retries, live status and per-document retrieval.
- Provider-agnostic AI layer (OpenAI or Gemini) with bring-your-own-key, AES-256-GCM encrypted keys, per-user rate limits and ownership checks on every route; Vitest and axe-core test coverage.

Symptoms Sense — AI Health Companion *Github*
FastAPI, OpenAI function calling, PostgreSQL, SQLAlchemy/Alembic, APScheduler, Next.js 14, TypeScript, Docker

- Conversational health assistant: describe symptoms, get pointed to the right specialist with urgency flags (information only, never a diagnosis), then book an appointment or set a medicine reminder in one chat.
- Tool-calling agent for doctor search, live availability, and booking, cancelling and rescheduling; **propose-then-confirm** design, so nothing is written until the user confirms.
- Full platform: patient, doctor and clinic accounts (JWT, Google sign-in), doctor directory with schedules and reviews, and scheduler-driven email medicine reminders; Dockerised.

CropSight — Smart Agriculture & Agro-Industry Management Platform *Live*
Next.js, Django, MongoDB, Google Earth Engine, TypeScript, Mapbox

- End-to-end farm and agri-project platform serving individual farms through to sugar and wheat mills, with web and Play Store mobile apps.
- GPS farm boundary mapping and field segmentation; remote-sensing land-cover and crop-health classification via Google Earth Engine.
- Geo-tagged task verification, vegetation and weather alerts, yield forecasting and stakeholder dashboards.

Kidney Stone Detection — CT image analysis, CNN vs SVM + HOG baseline (Gradio) *GitHub*

Research

GAN-Based Spatio-Temporal Fusion of Drone and Satellite Imagery for Precision Agriculture — *MS Thesis*

- GAN framework fusing drone imagery with 10 m Sentinel-2 data, using a spectral attention encoder and a multi-task segmentation head for weed, lodging and boundary detection.
- Trained on a self-collected dataset of 11,325 annotated patch pairs; **32.94 dB PSNR**, outperforming 14 competing methods by 5.7–7.1 points.
- Quantised to 8-bit for **CPU-only inference** — 230 MB, down from 850 MB, removing the GPU requirement.

Skills

AI & Agents: LangGraph, LangChain, MCP, Claude API, OpenAI API, Gemini API, RAG, Vector Search (pgvector, Pinecone), Agentic AI, Agent Orchestration
Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, GANs, Attention, Multi-Task Learning, Model Quantisation
Backend: Python, Node.js, FastAPI, Django, SQLAlchemy (async), PostgreSQL, Alembic, Redis/ARQ, REST APIs, JWT/OAuth2
Frontend: TypeScript, Next.js (App Router), React, Tailwind CSS, shadcn/ui, Zustand
Cloud & DevOps: AWS (EC2, RDS, S3, IAM, Secrets Manager), Docker, CI/CD, Ubuntu administration, Bash
Vision & Geospatial: OpenCV, YOLO, Segmentation, Google Earth Engine, Sentinel Hub, GIS, Mapbox
Other: C/C++, SQL, Git, Playwright, Stripe, Firebase

Education

MS Computer Science — COMSATS University Islamabad, Lahore *2024 – 2026*
BS Computer Science — COMSATS University Islamabad, Lahore *2020 – 2024*

Additional

Certifications: AI Developer (PIAIC) | ML Models in Microsoft Azure | Cloud Computing | Front-End Apps with React
Languages: English (Fluent) | Urdu (Native) | Punjabi (Bilingual)
Volunteering: TEDx COMSATS Lahore 2024 — Web & IT team; built the event website.